

Global Comparison Report

Monte Carlo Strategy Paired-Difference Analytics

GENERATED	MONTE CARLO RUNS	STRATEGIES TESTED
10 May 2026 16:41	40	2
METHOD	UNIT	REPORT TYPE
Discrete-Event Simulation (DES)	Haemodialysis Centre	Comparative Performance Analysis

Abstract

This report presents the results of a Monte Carlo discrete-event simulation study evaluating scheduling strategies for a haemodialysis unit. Stochastic variability in patient volume, machine availability, session duration, and nursing resources is modelled across 40 independent runs per strategy. Distributional summaries, tail-risk metrics (p90 / p99), and head-to-head strategy comparisons are provided to support operational decision-making.

Keywords: Monte Carlo | Discrete-Event Simulation | Haemodialysis | Scheduling | Tail-Risk

1. Simulation Setup

The parameters below fully specify the stochastic environment sampled during each Monte Carlo iteration. All stochastic quantities are drawn independently per run using a seeded Mersenne Twister PRNG.

TEMPORAL BOUNDS

Parameter	Value	Unit	Description
Shift Duration	360	min	Hard wall -- 6 h total shift
Nominal Session Duration	210 - 240	min	Prescribed per-patient session (~4 h target)
Min Viable Session	180	min	Sessions shorter than this are marked failed (3 h)
Machine Cooldown	60	min	Mandatory inter-session machine rest period

RESOURCE RANGES (uniform draw per iteration)

Parameter	Value	Unit	Description
Total Machines	15 - 20	units	Physical dialysis stations available
Patient Volume	15 - 20	patients	Demand sampled each shift
Nursing Staff	2 - 4	nurses	Concurrent supervision capacity
Session Duration	210 - 240	min	Per-patient dialysis time
Machine Ready Delay	0 - 90	min	Post-cooldown preparation lag

STOCHASTIC EVENT GENERATORS

Parameter	Value	Unit	Description
Patient Arrival Offset	Uniform[0, 60]	min	Arrival jitter relative to shift start
Setup Duration	Uniform[10, 20]	min	Machine preparation time per patient

PROBABILITY METRICS

Parameter	Value	Unit	Description
Machine Defect Probability	15.0%	--	Independent Bernoulli draw per machine per session

EXPERIMENT PARAMETERS

Parameter	Value	Unit	Description
Monte Carlo Iterations	40	runs	Independent replications per strategy

2. Global Statistical Summary

Per-strategy distributional metrics (mean +/- SD, p50, p90, p99) aggregated across the full Monte Carlo suite. The p90 and p99 columns serve as tail-risk indicators; lower is better.

FIFO

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	29.1 +/- 11.7	28.2	47.1	50.9

Max Wait Time (min)	52.2 +/- 17.7	48.0	75.5	91.0
Avg Session Time (min)	224.9 +/- 2.6	224.7	228.2	230.8

FIXED

Metric	Mean +/- SD	p50	p90	p99
Mean Wait Time (min)	58.6 +/- 17.0	55.5	80.0	88.0
Max Wait Time (min)	110.8 +/- 23.7	110.5	138.8	153.6
Avg Session Time (min)	221.6 +/- 4.5	222.4	227.6	228.5

3. Comparative Analytics

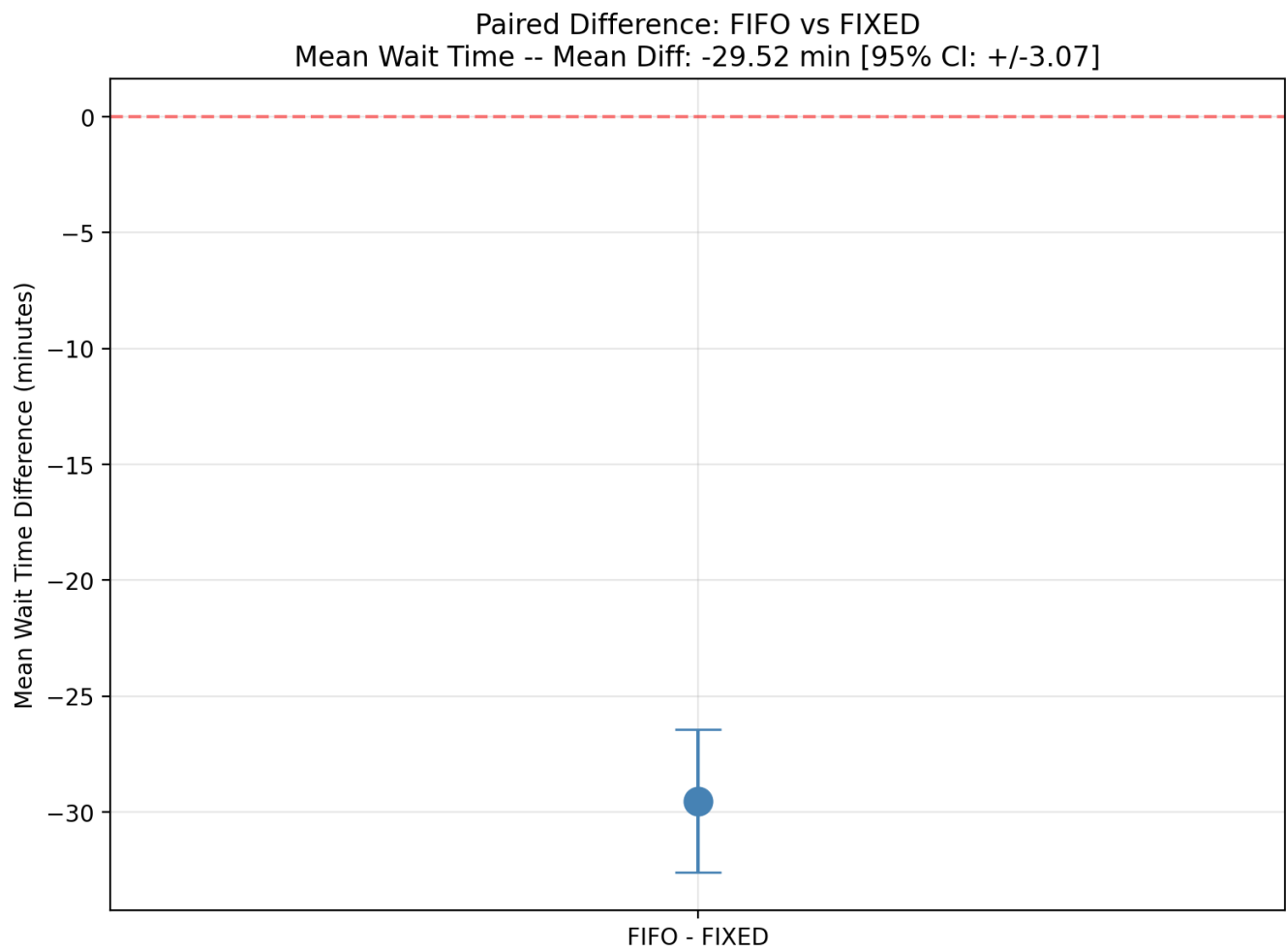


Figure 1. Paired-difference analysis of mean wait time across strategies.

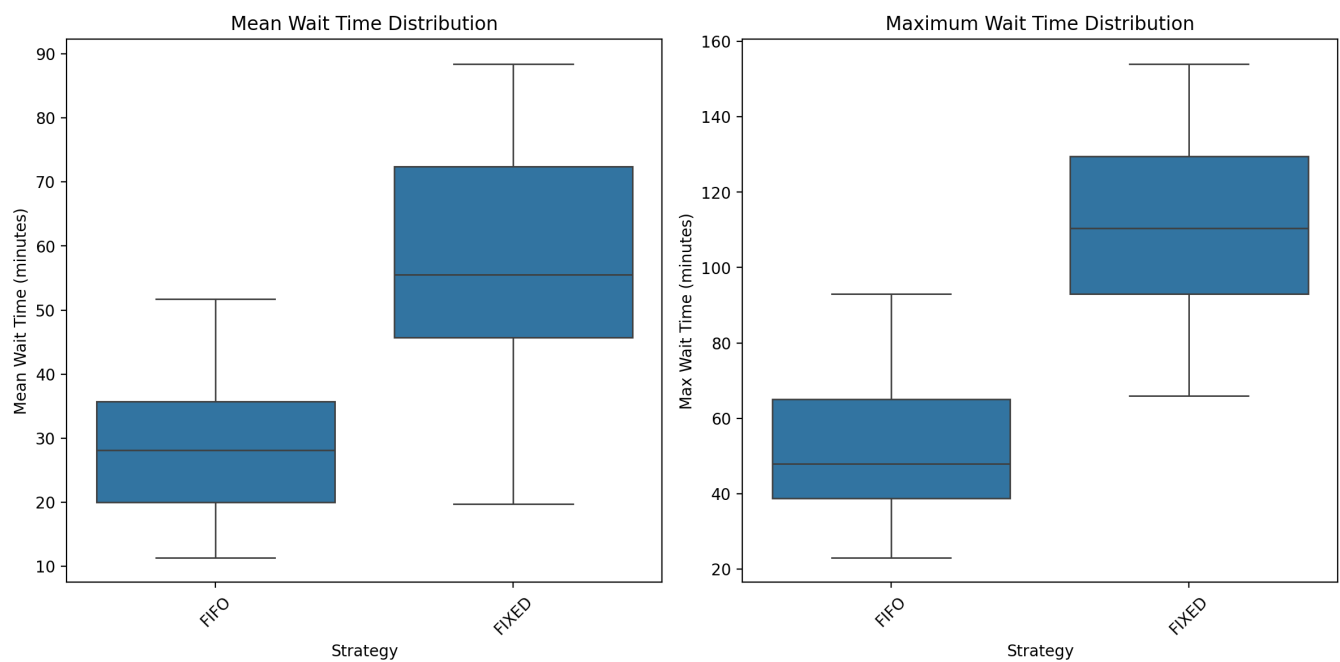


Figure 2. Wait-time distribution comparison across strategies.

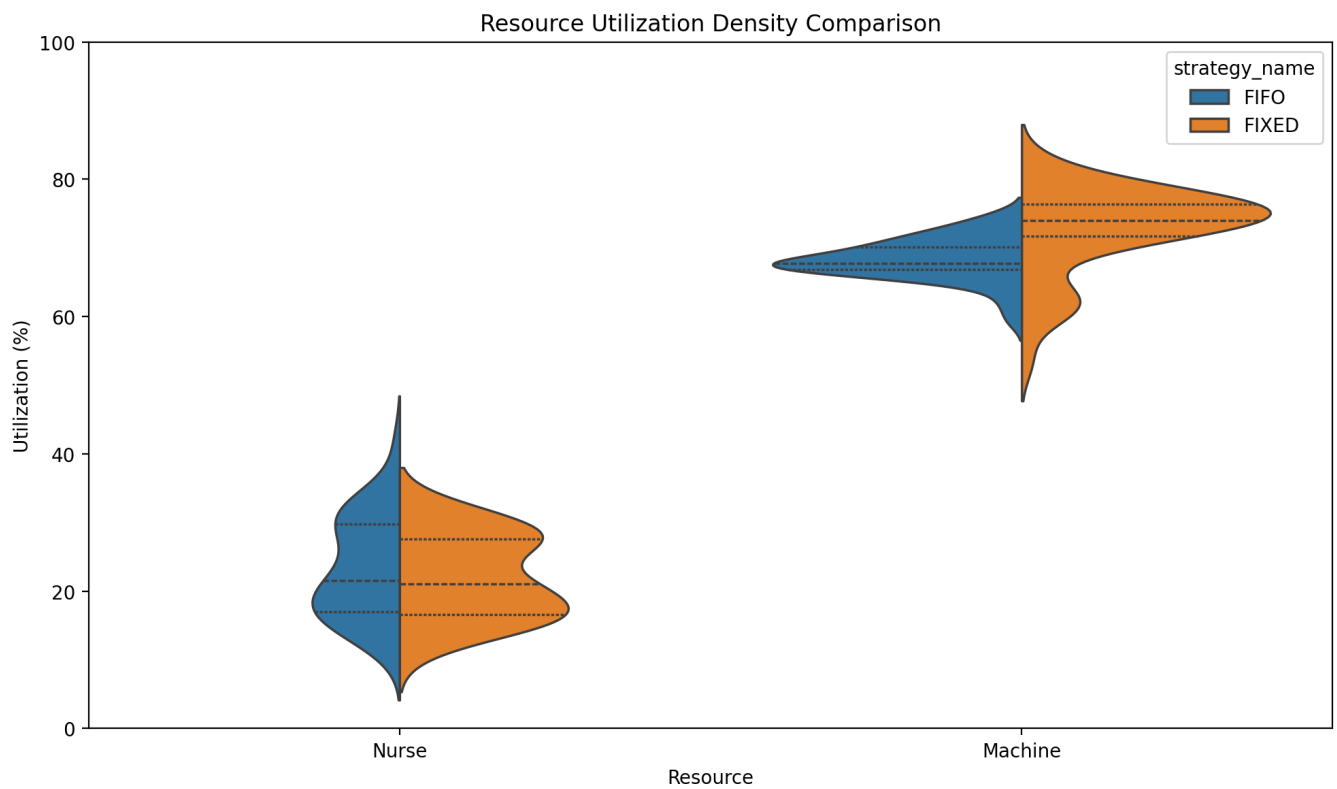


Figure 3. Utilisation violin plots across strategies.

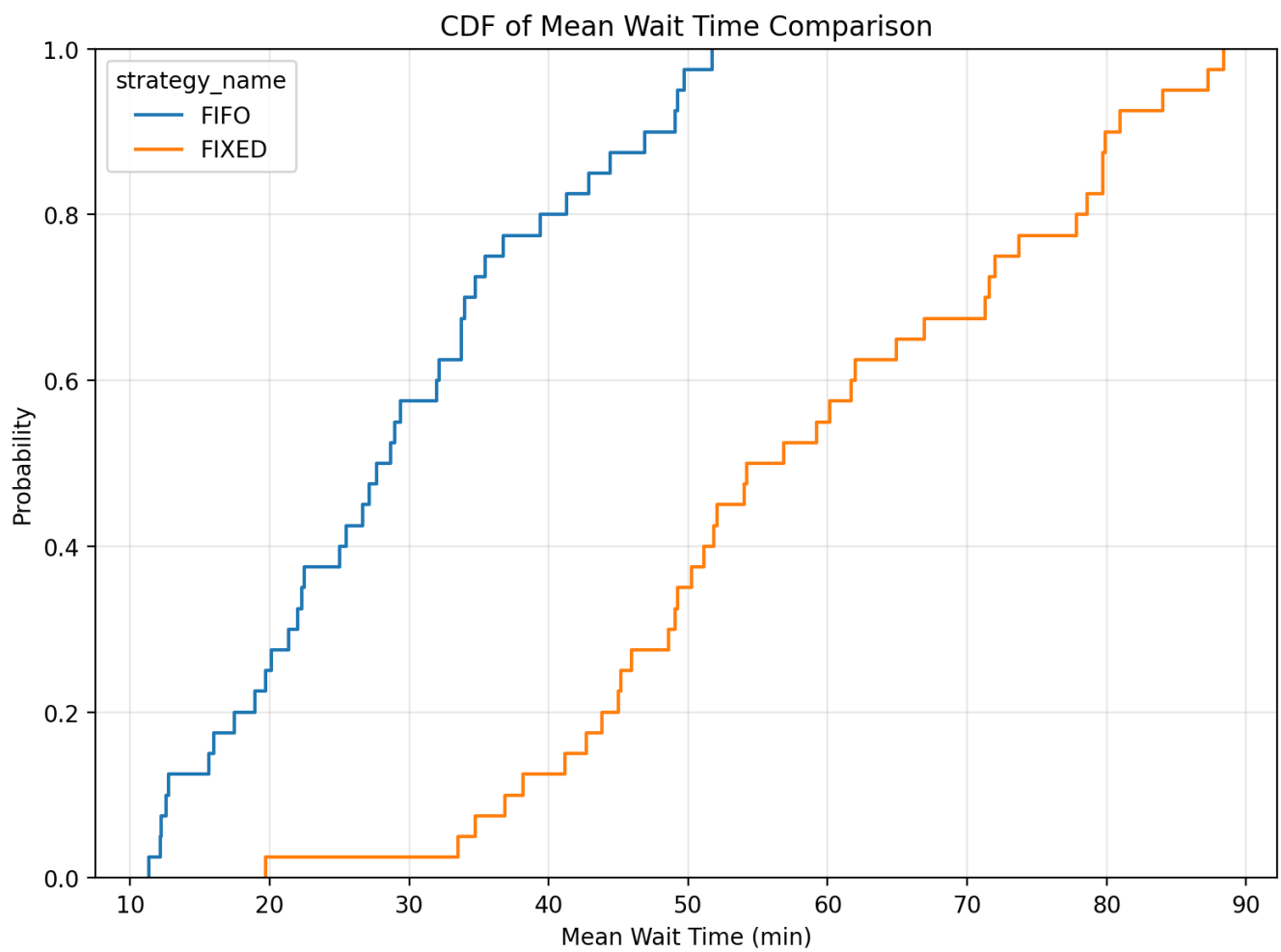


Figure 4. Empirical CDF of mean wait time -- all strategies.

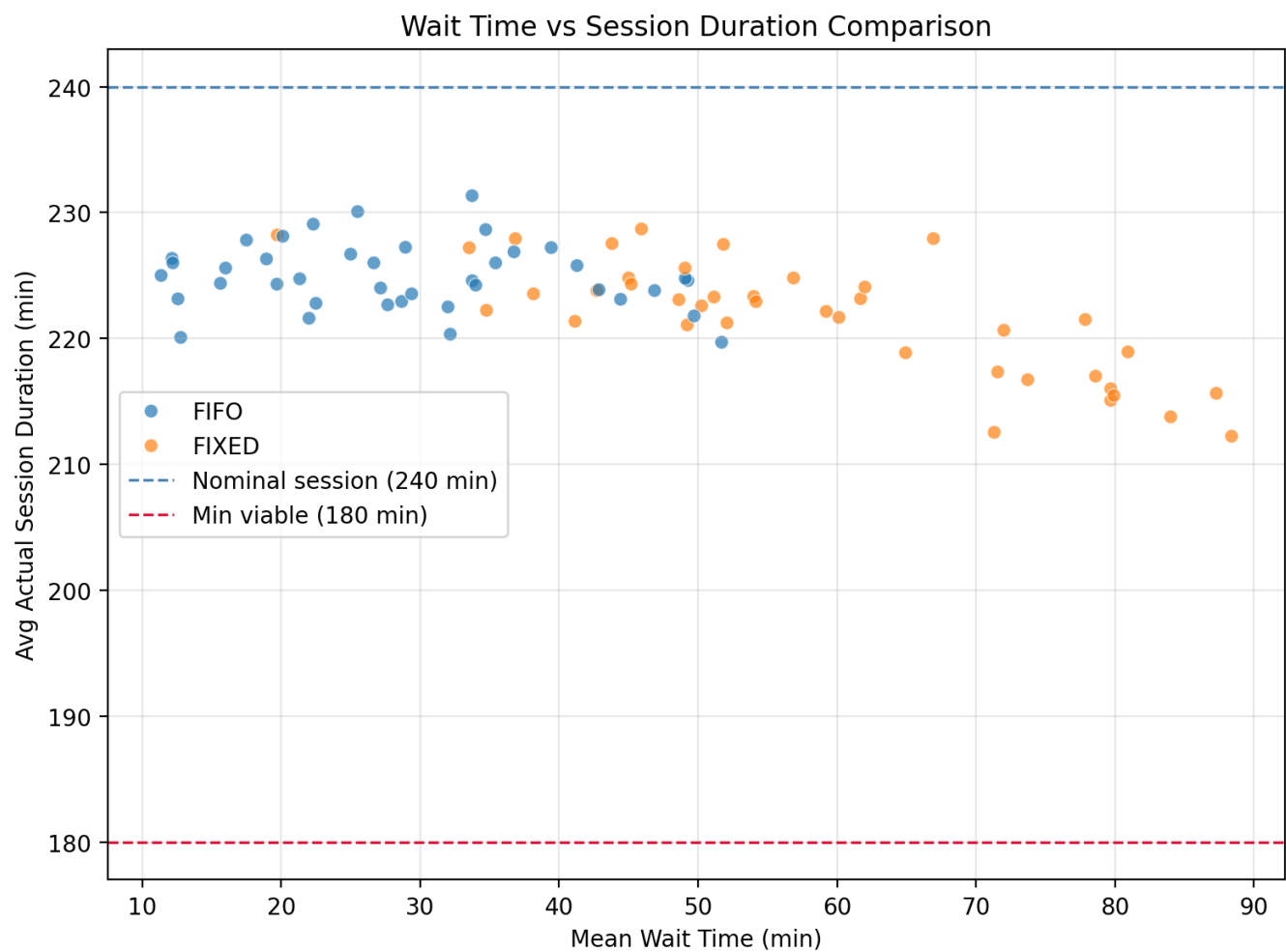


Figure 5. Wait time vs. avg session duration -- all strategies.

4. Edge-Case Stress Tests

Performance under extreme or auto-captured boundary conditions. Each card shows the full shift snapshot followed by per-strategy outcomes.

Scenario: Burst Arrival

Shift Snapshot

Active machines	20
Defective machines	0
Total machines	20
Nurses	2
Patients	20
Arrival window	0 - 0 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results

[PASS] FIFO | Patients: 20 | Failed: 0 | Truncated: 4

Wait (mean/max): 66.5 / 139.0 min | Avg session: 220 min | Nurse util: 41.4% | Machine util: 83.6%

[PASS] FIXED | Patients: 20 | Failed: 0 | Truncated: 4

Wait (mean/max): 66.5 / 139.0 min | Avg session: 220 min | Nurse util: 41.4% | Machine util: 83.6%

Scenario: Maintenance Disaster

Shift Snapshot

Active machines	15
Defective machines	12
Total machines	27
Nurses	4
Patients	20
Arrival window	1 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 17 | Truncated: 0

Wait (mean/max): 0.0 / 0.0 min | Avg session: 215 min | Nurse util: 3.1% | Machine util: 63.9%

[FAIL] FIXED | Patients: 20 | Failed: 17 | Truncated: 0

Wait (mean/max): 0.0 / 0.0 min | Avg session: 228 min | Nurse util: 2.6% | Machine util: 66.8%

Scenario: Marathon Shift

Shift Snapshot

Active machines	15
Defective machines	0
Total machines	15
Nurses	2
Patients	20
Arrival window	10 - 200 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	42

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 5 | Truncated: 5

Wait (mean/max): 0.0 / 0.0 min | Avg session: 230 min | Nurse util: 41.7% | Machine util: 69.4%

[FAIL] FIXED | Patients: 20 | Failed: 5 | Truncated: 5

Wait (mean/max): 0.0 / 0.0 min | Avg session: 230 min | Nurse util: 41.7% | Machine util: 69.4%

Scenario: Auto: seed-0 (FIFO)

Shift Snapshot

Active machines	13
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Defective machines	2
Total machines	15
Nurses	3
Patients	18
Arrival window	2 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	0

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 5 | Truncated: 0

Wait (mean/max): 25.0 / 48.0 min | Avg session: 227 min | Nurse util: 19.6% | Machine util: 67.9%

Scenario: Auto: seed-0 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	18
Arrival window	2 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	0

Strategy Results

[FAIL] FIXED | Patients: 18 | Failed: 5 | Truncated: 1

Wait (mean/max): 45.9 / 88.0 min | Avg session: 229 min | Nurse util: 18.1% | Machine util: 75.6%

Scenario: Auto: seed-1 (FIXED)**Shift Snapshot**

Active machines	17
Defective machines	1
Total machines	18
Nurses	4
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	1

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 1 | Truncated: 0

Wait (mean/max): 43.8 / 94.0 min | Avg session: 228 min | Nurse util: 15.2% | Machine util: 64.3%

Scenario: Auto: seed-2 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	2
Patients	15
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	2

Strategy Results

[FAIL] FIXED | Patients: 15 | Failed: 1 | Truncated: 7

Wait (mean/max): 73.7 / 131.0 min | Avg session: 217 min | Nurse util: 31.7% | Machine util: 74.7%

Scenario: Auto: seed-3 (FIXED)**Shift Snapshot**

Active machines	18
Defective machines	1
Total machines	19
Nurses	4
Patients	16
Arrival window	9 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	3

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 1 | Truncated: 0

Wait (mean/max): 45.0 / 89.0 min | Avg session: 225 min | Nurse util: 15.5% | Machine util: 60.4%

Scenario: Auto: seed-4 (FIFO)**Shift Snapshot**

Active machines	15
Defective machines	3
Total machines	18
Nurses	3
Patients	16
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	4

Strategy Results**[FAIL] FIFO | Patients: 16 | Failed: 1 | Truncated: 0**

Wait (mean/max): 28.7 / 38.0 min | Avg session: 223 min | Nurse util: 19.8% | Machine util: 66.5%

Scenario: Auto: seed-4 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	3
Total machines	18
Nurses	3
Patients	16
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	4

Strategy Results**[FAIL] FIXED | Patients: 16 | Failed: 3 | Truncated: 0**

Wait (mean/max): 48.6 / 96.0 min | Avg session: 223 min | Nurse util: 16.2% | Machine util: 62.0%

Scenario: Auto: seed-5 (FIFO)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	19
Arrival window	0 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	5

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0**

Wait (mean/max): 26.7 / 47.0 min | Avg session: 226 min | Nurse util: 19.8% | Machine util: 67.2%

Scenario: Auto: seed-5 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	19
Arrival window	0 - 56 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	5

Strategy Results

[FAIL] FIXED | Patients: 19 | Failed: 4 | Truncated: 1

Wait (mean/max): 56.9 / 111.0 min | Avg session: 225 min | Nurse util: 19.9% | Machine util: 75.7%

Scenario: Auto: seed-6 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	6

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0

Wait (mean/max): 33.7 / 65.0 min | Avg session: 231 min | Nurse util: 31.5% | Machine util: 72.2%

Scenario: Auto: seed-6 (FIXED)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	6

Strategy Results

[FAIL] FIXED | Patients: 19 | Failed: 4 | Truncated: 7

Wait (mean/max): 78.6 / 146.0 min | Avg session: 217 min | Nurse util: 29.9% | Machine util: 81.0%

Scenario: Auto: seed-7 (FIFO)

Shift Snapshot

Active machines	13
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Defective machines	2
Total machines	15
Nurses	2
Patients	17
Arrival window	2 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	7

Strategy Results

[FAIL] FIFO | Patients: 17 | Failed: 4 | Truncated: 0

Wait (mean/max): 51.7 / 82.0 min | Avg session: 220 min | Nurse util: 29.0% | Machine util: 69.9%

Scenario: Auto: seed-7 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	2
Patients	17
Arrival window	2 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	7

Strategy Results

[FAIL] FIXED | Patients: 17 | Failed: 4 | Truncated: 6

Wait (mean/max): 88.4 / 148.0 min | Avg session: 212 min | Nurse util: 29.0% | Machine util: 78.6%

Scenario: Auto: seed-8 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	3
Total machines	16
Nurses	3
Patients	16
Arrival window	1 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	8

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 3 | Truncated: 0

Wait (mean/max): 29.4 / 50.0 min | Avg session: 224 min | Nurse util: 17.1% | Machine util: 66.5%

Scenario: Auto: seed-8 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	3
Total machines	16
Nurses	3
Patients	16
Arrival window	1 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	8

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 3 | Truncated: 0

Wait (mean/max): 52.1 / 100.0 min | Avg session: 221 min | Nurse util: 16.9% | Machine util: 72.2%

Scenario: Auto: seed-9 (FIFO)**Shift Snapshot**

Active machines	14
Defective machines	3
Total machines	17
Nurses	4
Patients	18
Arrival window	5 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	9

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 4 | Truncated: 0

Wait (mean/max): 12.1 / 34.0 min | Avg session: 226 min | Nurse util: 14.1% | Machine util: 67.6%

Scenario: Auto: seed-9 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	3
Total machines	17
Nurses	4
Patients	18
Arrival window	5 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	9

Strategy Results**[FAIL] FIXED | Patients: 18 | Failed: 4 | Truncated: 0**

Wait (mean/max): 19.7 / 73.0 min | Avg session: 228 min | Nurse util: 14.9% | Machine util: 70.7%

Scenario: Auto: seed-10 (FIFO)**Shift Snapshot**

Active machines	15
Defective machines	5
Total machines	20
Nurses	2
Patients	19
Arrival window	0 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	10

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 4 | Truncated: 0**

Wait (mean/max): 49.3 / 88.0 min | Avg session: 225 min | Nurse util: 31.0% | Machine util: 71.1%

Scenario: Auto: seed-10 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	5
Total machines	20
Nurses	2
Patients	19
Arrival window	0 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	10

Strategy Results**[FAIL] FIXED | Patients: 19 | Failed: 5 | Truncated: 7**

Wait (mean/max): 87.3 / 138.0 min | Avg session: 216 min | Nurse util: 28.1% | Machine util: 75.5%

Scenario: Auto: seed-11 (FIFO)**Shift Snapshot**

Active machines	14
Defective machines	1
Total machines	15
Nurses	4
Patients	18
Arrival window	2 - 54 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	11

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 4 | Truncated: 0

Wait (mean/max): 12.6 / 51.0 min | Avg session: 223 min | Nurse util: 15.7% | Machine util: 66.6%

Scenario: Auto: seed-11 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	1
Total machines	15
Nurses	4
Patients	18
Arrival window	2 - 54 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	11

Strategy Results

[FAIL] FIXED | Patients: 18 | Failed: 4 | Truncated: 0

Wait (mean/max): 51.1 / 100.0 min | Avg session: 223 min | Nurse util: 15.6% | Machine util: 75.1%

Scenario: Auto: seed-13 (FIFO)

Shift Snapshot

Active machines	15
Defective machines	2
Total machines	17
Nurses	3
Patients	17
Arrival window	4 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	13

Strategy Results

[FAIL] FIFO | Patients: 17 | Failed: 2 | Truncated: 0

Wait (mean/max): 21.3 / 41.0 min | Avg session: 225 min | Nurse util: 21.2% | Machine util: 67.3%

Scenario: Auto: seed-13 (FIXED)

Shift Snapshot

Active machines	15
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Defective machines	2
Total machines	17
Nurses	3
Patients	17
Arrival window	4 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	13

Strategy Results

[FAIL] FIXED | Patients: 17 | Failed: 2 | Truncated: 3

Wait (mean/max): 54.0 / 99.0 min | Avg session: 223 min | Nurse util: 21.0% | Machine util: 74.1%

Scenario: Auto: seed-14 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	4
Patients	15
Arrival window	14 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	14

Strategy Results

[FAIL] FIFO | Patients: 15 | Failed: 2 | Truncated: 0

Wait (mean/max): 12.8 / 36.0 min | Avg session: 220 min | Nurse util: 12.8% | Machine util: 65.1%

Scenario: Auto: seed-14 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	4
Patients	15
Arrival window	14 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	14

Strategy Results

[FAIL] FIXED | Patients: 15 | Failed: 2 | Truncated: 0

Wait (mean/max): 34.8 / 66.0 min | Avg session: 222 min | Nurse util: 13.3% | Machine util: 71.4%

Scenario: Auto: seed-15 (FIXED)**Shift Snapshot**

Active machines	16
Defective machines	2
Total machines	18
Nurses	2
Patients	16
Arrival window	3 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	15

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 3 | Truncated: 4

Wait (mean/max): 71.3 / 118.0 min | Avg session: 213 min | Nurse util: 27.8% | Machine util: 62.0%

Scenario: Auto: seed-16 (FIXED)**Shift Snapshot**

Active machines	17
Defective machines	3
Total machines	20
Nurses	3
Patients	17
Arrival window	1 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	16

Strategy Results

[FAIL] FIXED | Patients: 17 | Failed: 3 | Truncated: 0

Wait (mean/max): 42.7 / 76.0 min | Avg session: 224 min | Nurse util: 19.8% | Machine util: 61.2%

Scenario: Auto: seed-17 (FIFO)**Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	3
Patients	19
Arrival window	1 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	17

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 5 | Truncated: 0**

Wait (mean/max): 11.4 / 31.0 min | Avg session: 225 min | Nurse util: 19.9% | Machine util: 67.1%

Scenario: Auto: seed-17 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	3
Patients	19
Arrival window	1 - 56 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	17

Strategy Results**[FAIL] FIXED | Patients: 19 | Failed: 5 | Truncated: 1**

Wait (mean/max): 41.1 / 89.0 min | Avg session: 221 min | Nurse util: 20.7% | Machine util: 73.2%

Scenario: Auto: seed-18 (FIFO)**Shift Snapshot**

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	18

Strategy Results**[FAIL] FIFO | Patients: 16 | Failed: 2 | Truncated: 1**

Wait (mean/max): 41.3 / 75.0 min | Avg session: 226 min | Nurse util: 30.7% | Machine util: 70.1%

Scenario: Auto: seed-18 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	16
Arrival window	0 - 60 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	18

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 3 | Truncated: 6

Wait (mean/max): 79.7 / 154.0 min | Avg session: 215 min | Nurse util: 29.2% | Machine util: 73.2%

Scenario: Auto: seed-19 (FIFO)**Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	2
Patients	20
Arrival window	6 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	19

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 6 | Truncated: 0

Wait (mean/max): 35.4 / 70.0 min | Avg session: 226 min | Nurse util: 29.4% | Machine util: 69.1%

Scenario: Auto: seed-19 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	5
Total machines	19
Nurses	2
Patients	20
Arrival window	6 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	19

Strategy Results

[FAIL] FIXED | Patients: 20 | Failed: 7 | Truncated: 7

Wait (mean/max): 84.0 / 136.0 min | Avg session: 214 min | Nurse util: 26.8% | Machine util: 74.0%

Scenario: Auto: seed-20 (FIFO)**Shift Snapshot**

Active machines	12
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Defective machines	4
Total machines	16
Nurses	4
Patients	20
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	20

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 8 | Truncated: 0

Wait (mean/max): 33.8 / 56.0 min | Avg session: 225 min | Nurse util: 11.2% | Machine util: 66.1%

Scenario: Auto: seed-20 (FIXED)

Shift Snapshot

Active machines	12
Defective machines	4
Total machines	16
Nurses	4
Patients	20
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	20

Strategy Results

[FAIL] FIXED | Patients: 20 | Failed: 8 | Truncated: 0

Wait (mean/max): 54.2 / 100.0 min | Avg session: 223 min | Nurse util: 11.1% | Machine util: 73.9%

Scenario: Auto: seed-21 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	21

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 3 | Truncated: 0

Wait (mean/max): 22.3 / 37.0 min | Avg session: 229 min | Nurse util: 16.0% | Machine util: 67.6%

Scenario: Auto: seed-21 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	2
Total machines	15
Nurses	3
Patients	16
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	21

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 3 | Truncated: 0

Wait (mean/max): 36.8 / 70.0 min | Avg session: 228 min | Nurse util: 16.9% | Machine util: 72.0%

Scenario: Auto: seed-22 (FIXED)**Shift Snapshot**

Active machines	16
Defective machines	2
Total machines	18
Nurses	2
Patients	16
Arrival window	1 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	22

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 2 | Truncated: 5

Wait (mean/max): 72.0 / 133.0 min | Avg session: 221 min | Nurse util: 28.2% | Machine util: 68.7%

Scenario: Auto: seed-23 (FIFO)**Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	2
Patients	17
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	23

Strategy Results**[FAIL] FIFO | Patients: 17 | Failed: 4 | Truncated: 0**

Wait (mean/max): 49.1 / 75.0 min | Avg session: 225 min | Nurse util: 28.3% | Machine util: 69.8%

Scenario: Auto: seed-23 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	2
Patients	17
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	23

Strategy Results**[FAIL] FIXED | Patients: 17 | Failed: 4 | Truncated: 4**

Wait (mean/max): 79.7 / 135.0 min | Avg session: 216 min | Nurse util: 26.5% | Machine util: 77.6%

Scenario: Auto: seed-24 (FIFO)**Shift Snapshot**

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	20
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	24

Strategy Results**[FAIL] FIFO | Patients: 20 | Failed: 3 | Truncated: 0**

Wait (mean/max): 28.9 / 42.0 min | Avg session: 227 min | Nurse util: 24.9% | Machine util: 71.3%

Scenario: Auto: seed-24 (FIXED)**Shift Snapshot**

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	20
Arrival window	1 - 59 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	24

Strategy Results

[FAIL] FIXED | Patients: 20 | Failed: 3 | Truncated: 1

Wait (mean/max): 49.1 / 108.0 min | Avg session: 226 min | Nurse util: 23.9% | Machine util: 76.9%

Scenario: Auto: seed-25 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	2
Total machines	16
Nurses	2
Patients	18
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	25

Strategy Results

[FAIL] FIFO | Patients: 18 | Failed: 4 | Truncated: 0

Wait (mean/max): 39.4 / 65.0 min | Avg session: 227 min | Nurse util: 31.8% | Machine util: 72.3%

Scenario: Auto: seed-25 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	2
Total machines	16
Nurses	2
Patients	18
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	25

Strategy Results

[FAIL] FIXED | Patients: 18 | Failed: 4 | Truncated: 5

Wait (mean/max): 64.9 / 133.0 min | Avg session: 219 min | Nurse util: 31.8% | Machine util: 76.6%

Scenario: Auto: seed-26 (FIFO)

Shift Snapshot

Active machines	13
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Defective machines	4
Total machines	17
Nurses	2
Patients	20
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	26

Strategy Results**[FAIL] FIFO | Patients: 20 | Failed: 7 | Truncated: 0**

Wait (mean/max): 27.2 / 53.0 min | Avg session: 224 min | Nurse util: 26.8% | Machine util: 68.8%

Scenario: Auto: seed-26 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	4
Total machines	17
Nurses	2
Patients	20
Arrival window	0 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	26

Strategy Results**[FAIL] FIXED | Patients: 20 | Failed: 7 | Truncated: 2**

Wait (mean/max): 61.7 / 127.0 min | Avg session: 223 min | Nurse util: 26.2% | Machine util: 77.1%

Scenario: Auto: seed-27 (FIFO)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	20
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	27

Strategy Results**[FAIL] FIFO | Patients: 20 | Failed: 5 | Truncated: 0**

Wait (mean/max): 12.2 / 23.0 min | Avg session: 226 min | Nurse util: 18.6% | Machine util: 66.8%

Scenario: Auto: seed-27 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	1
Total machines	16
Nurses	3
Patients	20
Arrival window	4 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	27

Strategy Results

[FAIL] FIXED | Patients: 20 | Failed: 5 | Truncated: 0

Wait (mean/max): 33.5 / 83.0 min | Avg session: 227 min | Nurse util: 19.7% | Machine util: 73.8%

Scenario: Auto: seed-28 (FIXED)**Shift Snapshot**

Active machines	16
Defective machines	1
Total machines	17
Nurses	4
Patients	15
Arrival window	6 - 58 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	28

Strategy Results

[FAIL] FIXED | Patients: 15 | Failed: 1 | Truncated: 0

Wait (mean/max): 49.2 / 99.0 min | Avg session: 221 min | Nurse util: 15.4% | Machine util: 62.4%

Scenario: Auto: seed-29 (FIXED)**Shift Snapshot**

Active machines	19
Defective machines	1
Total machines	20
Nurses	2
Patients	19
Arrival window	6 - 53 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	29

Strategy Results**[FAIL] FIXED | Patients: 19 | Failed: 6 | Truncated: 6**

Wait (mean/max): 79.9 / 126.0 min | Avg session: 215 min | Nurse util: 27.5% | Machine util: 53.7%

Scenario: Auto: seed-30 (FIFO)**Shift Snapshot**

Active machines	16
Defective machines	1
Total machines	17
Nurses	3
Patients	19
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	30

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 3 | Truncated: 0**

Wait (mean/max): 18.9 / 39.0 min | Avg session: 226 min | Nurse util: 22.0% | Machine util: 68.0%

Scenario: Auto: seed-30 (FIXED)**Shift Snapshot**

Active machines	16
Defective machines	1
Total machines	17
Nurses	3
Patients	19
Arrival window	1 - 59 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	30

Strategy Results**[FAIL] FIXED | Patients: 19 | Failed: 3 | Truncated: 2**

Wait (mean/max): 45.2 / 113.0 min | Avg session: 224 min | Nurse util: 21.9% | Machine util: 73.8%

Scenario: Auto: seed-31 (FIFO)**Shift Snapshot**

Active machines	12
Defective machines	3
Total machines	15
Nurses	3
Patients	15
Arrival window	3 - 60 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	31

Strategy Results

[FAIL] FIFO | Patients: 15 | Failed: 3 | Truncated: 0

Wait (mean/max): 32.2 / 48.0 min | Avg session: 220 min | Nurse util: 15.4% | Machine util: 65.2%

Scenario: Auto: seed-31 (FIXED)**Shift Snapshot**

Active machines	12
Defective machines	3
Total machines	15
Nurses	3
Patients	15
Arrival window	3 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	31

Strategy Results

[FAIL] FIXED | Patients: 15 | Failed: 3 | Truncated: 0

Wait (mean/max): 50.2 / 84.0 min | Avg session: 223 min | Nurse util: 15.8% | Machine util: 71.9%

Scenario: Auto: seed-32 (FIFO)**Shift Snapshot**

Active machines	14
Defective machines	1
Total machines	15
Nurses	2
Patients	15
Arrival window	0 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	32

Strategy Results

[FAIL] FIFO | Patients: 15 | Failed: 1 | Truncated: 0

Wait (mean/max): 22.5 / 39.0 min | Avg session: 223 min | Nurse util: 27.1% | Machine util: 70.2%

Scenario: Auto: seed-32 (FIXED)**Shift Snapshot**

Active machines	14
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Defective machines	1
Total machines	15
Nurses	2
Patients	15
Arrival window	0 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	32

Strategy Results

[FAIL] FIXED | Patients: 15 | Failed: 1 | Truncated: 5

Wait (mean/max): 80.9 / 153.0 min | Avg session: 219 min | Nurse util: 27.5% | Machine util: 82.0%

Scenario: Auto: seed-33 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	19
Arrival window	3 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	33

Strategy Results

[FAIL] FIFO | Patients: 19 | Failed: 5 | Truncated: 1

Wait (mean/max): 25.5 / 59.0 min | Avg session: 230 min | Nurse util: 31.5% | Machine util: 72.2%

Scenario: Auto: seed-33 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	3
Total machines	17
Nurses	2
Patients	19
Arrival window	3 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	33

Strategy Results

[FAIL] FIXED | Patients: 19 | Failed: 5 | Truncated: 6

Wait (mean/max): 59.2 / 122.0 min | Avg session: 222 min | Nurse util: 30.1% | Machine util: 78.6%

Scenario: Auto: seed-34 (FIFO)**Shift Snapshot**

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	19
Arrival window	3 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	34

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 2 | Truncated: 0**

Wait (mean/max): 20.1 / 34.0 min | Avg session: 228 min | Nurse util: 21.9% | Machine util: 68.1%

Scenario: Auto: seed-34 (FIXED)**Shift Snapshot**

Active machines	17
Defective machines	1
Total machines	18
Nurses	3
Patients	19
Arrival window	3 - 60 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	34

Strategy Results**[FAIL] FIXED | Patients: 19 | Failed: 2 | Truncated: 1**

Wait (mean/max): 51.8 / 103.0 min | Avg session: 227 min | Nurse util: 21.1% | Machine util: 76.0%

Scenario: Auto: seed-35 (FIFO)**Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	3
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	35

Strategy Results**[FAIL] FIFO | Patients: 19 | Failed: 6 | Truncated: 0**

Wait (mean/max): 34.0 / 59.0 min | Avg session: 224 min | Nurse util: 19.2% | Machine util: 67.0%

Scenario: Auto: seed-35 (FIXED)**Shift Snapshot**

Active machines	13
Defective machines	5
Total machines	18
Nurses	3
Patients	19
Arrival window	0 - 55 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	35

Strategy Results**[FAIL] FIXED | Patients: 19 | Failed: 6 | Truncated: 2**

Wait (mean/max): 62.0 / 110.0 min | Avg session: 224 min | Nurse util: 19.1% | Machine util: 76.0%

Scenario: Auto: seed-36 (FIFO)**Shift Snapshot**

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	17
Arrival window	0 - 57 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	36

Strategy Results**[FAIL] FIFO | Patients: 17 | Failed: 2 | Truncated: 0**

Wait (mean/max): 42.9 / 70.0 min | Avg session: 224 min | Nurse util: 32.4% | Machine util: 74.3%

Scenario: Auto: seed-36 (FIXED)**Shift Snapshot**

Active machines	15
Defective machines	2
Total machines	17
Nurses	2
Patients	17
Arrival window	0 - 57 min

Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	36

Strategy Results

[FAIL] FIXED | Patients: 17 | Failed: 2 | Truncated: 3

Wait (mean/max): 60.1 / 124.0 min | Avg session: 222 min | Nurse util: 32.2% | Machine util: 77.1%

Scenario: Auto: seed-37 (FIFO)

Shift Snapshot

Active machines	13
Defective machines	3
Total machines	16
Nurses	4
Patients	20
Arrival window	2 - 53 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	37

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 7 | Truncated: 0

Wait (mean/max): 19.7 / 46.0 min | Avg session: 224 min | Nurse util: 14.0% | Machine util: 66.8%

Scenario: Auto: seed-37 (FIXED)

Shift Snapshot

Active machines	13
Defective machines	3
Total machines	16
Nurses	4
Patients	20
Arrival window	2 - 53 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	37

Strategy Results

[FAIL] FIXED | Patients: 20 | Failed: 7 | Truncated: 0

Wait (mean/max): 38.2 / 90.0 min | Avg session: 224 min | Nurse util: 15.3% | Machine util: 73.3%

Scenario: Auto: seed-38 (FIFO)

Shift Snapshot

Active machines	14
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Defective machines	4
Total machines	18
Nurses	3
Patients	20
Arrival window	0 - 49 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	38

Strategy Results

[FAIL] FIFO | Patients: 20 | Failed: 6 | Truncated: 0

Wait (mean/max): 49.7 / 69.0 min | Avg session: 222 min | Nurse util: 20.6% | Machine util: 67.3%

Scenario: Auto: seed-38 (FIXED)

Shift Snapshot

Active machines	14
Defective machines	4
Total machines	18
Nurses	3
Patients	20
Arrival window	0 - 49 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	38

Strategy Results

[FAIL] FIXED | Patients: 20 | Failed: 6 | Truncated: 3

Wait (mean/max): 77.9 / 129.0 min | Avg session: 222 min | Nurse util: 21.1% | Machine util: 76.0%

Scenario: Auto: seed-39 (FIFO)

Shift Snapshot

Active machines	14
Defective machines	2
Total machines	16
Nurses	3
Patients	16
Arrival window	0 - 52 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	39

Strategy Results

[FAIL] FIFO | Patients: 16 | Failed: 2 | Truncated: 0

Wait (mean/max): 34.7 / 45.0 min | Avg session: 229 min | Nurse util: 17.1% | Machine util: 67.4%

Scenario: Auto: seed-39 (FIXED)**Shift Snapshot**

Active machines	14
Defective machines	2
Total machines	16
Nurses	3
Patients	16
Arrival window	0 - 52 min
Shift duration	360 min
Min viable session	180 min
Machine cooldown	60 min
Scenario seed	39

Strategy Results

[FAIL] FIXED | Patients: 16 | Failed: 2 | Truncated: 1

Wait (mean/max): 66.9 / 113.0 min | Avg session: 228 min | Nurse util: 16.7% | Machine util: 76.3%